

The empirical recurrence for OEIS A224203, recovered from its tail

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Abstract

OEIS A224203 records “Empirical recurrence of order 41” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 610$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 49$. Its last coefficient is $c_{41} = 68097422131200 \neq 0$, so no further tail satisfies anything shorter; any order-41 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A224203 is “Number of $n \times 7$ 0..3 arrays with rows unimodal and antidiagonals nondecreasing.”. It has offset 1 and begins

1163, 180821, 11446096, 475218035, 17178450136, 618703866073, 22835261226043, 856178295327574, ...

The entry states:

Empirical recurrence of order 41 (see link above)

As of the “Last modified” line on the live entry (Oct 03 2025 23:19:16, revision 10) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 610$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 610$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 41: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 41.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1220$ exact terms from $n = 8$ on, it returns order 41 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{41} c_i a(n-i),$$

c_1	120	c_{12}	−2683978430932	c_{23}	1050524403777918000	c_{34}	2921299068148297728
c_2	−6060	c_{13}	10780238513518	c_{24}	2113714471975471104	c_{35}	856946059741420288
c_3	173316	c_{14}	−32948873206590	c_{25}	5481174460069822848	c_{36}	−327218428029860352
c_4	−3172287	c_{15}	77959459244560	c_{26}	10535537672809153264	c_{37}	−16665963801149440
c_5	39656548	c_{16}	−153434141519476	c_{27}	27173919094636341472	c_{38}	−24912329585756160
c_6	−352443358	c_{17}	1059262494852896	c_{28}	47704013358770735568	c_{39}	4712255288340480
c_7	2297379936	c_{18}	−2694841466898608	c_{29}	67212912534846684448	c_{40}	807198452121600
c_8	−11353877196	c_{19}	12219509145685508	c_{30}	62342889733184689824	c_{41}	68097422131200
c_9	45067977530	c_{20}	11985467868644368	c_{31}	49243871592902490976		
c_{10}	−163575910944	c_{21}	94047496662754280	c_{32}	27263322225301218304		
c_{11}	627001572604	c_{22}	94835829721618640	c_{33}	12918679041553050240		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 49$, and it is the recurrence OEIS A224203 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{41} - c_1 x^{40} - \dots - c_{41}$. Its constant term is $-c_{41} = -68097422131200$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 41 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 41, so $\deg q = 41$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 41, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 41, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 68097422131200.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-41 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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